

Assignment-1

Data Communication and Computer Networks (CS221)

Submit on or before 5th August

1. Explain the OSI (Open Systems Interconnection) reference model with a neat block diagram. Describe the functions of each of its seven layers and discuss the significance of the layered architecture in data communication.
2. What are transmission impairments? Explain attenuation, distortion, and noise with suitable diagrams. Derive the expression for Signal-to-Noise Ratio (SNR) in decibels and discuss its significance in data communication.
3. A communication channel has a bandwidth of 5 kHz and a signal-to-noise ratio of 40 dB. Determine: (a) the Shannon channel capacity, (b) the maximum bit rate using Nyquist criterion for an 8-level signaling scheme.
4. A digital communication system transmits data at a bit rate of 240 kbps. One data element is represented by one signal element. Assuming the average case factor $C=1/2$. (a) Calculate the signal rate (baud rate). (b) Determine the minimum bandwidth required for transmission.
5. Encode the data sequence '1011001' using Unipolar NRZ, Polar NRZ-L, NRZ-I, Manchester, Differential Manchester, Bipolar AMI and Pseudoternary coding techniques and draw the corresponding waveform.